

Year 3 Group Design Project

Shockingly Fast Electric Aircraft



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Abstract

The aim of the shockingly fast aircraft project was to develop an electric aerobatic aircraft in order to demonstrate the feasibility of using an electric motor. In this report, the challenges brought by this type of propulsion on the aerodynamics and more precisely on the wing were analysed. In order to quickly iterate through the various design a VLM method enhanced with viscous effects was developed. This method proved to be within a 5% error from wind tunnel data. In addition a selection process for the aerofoils using a simulation of the trajectory was developed. In order to decrease the overall weight of the batteries, winglets were fitted to the wing. The winglets proved to be a valuable addition to the wing as an increase in the lift to drag ratio of 10% was obtained at an angle of attack of 8° while no reduction were observed at lower angle.

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Nomenclature

λ	Taper ratio, $\lambda = c_{tip}/c_{root}$
P_a	Available power
P_r	Required power
P_s	Specific excess power
$\frac{dh}{dt}$	Climb or descent rate
a	3D lift curve slope
a_0	2D Lift curve slope
AR	Aspect Ratio, $AR = b^2/S$
b	Span
C_d	2D drag coefficient
C_D	3D drag coefficient
C_l	2D lift coefficient
C_L	3D lift coefficient
C_M	Moment coefficient
C_{D_0}	Zero-lift drag
C_{D_i}	Induced drag
S_h	Area of the horizontal tail
S_{ref}	Reference area of the wing
T	Thrust (N)
T	Thrust (N)
T/W	Thrust to weight ratio
W/S	Wing loading Nm^2
e	Oswald efficiency
n	Load factor
Re	Reynolds number
s	Semi-span of the wing
t/c	thickness to chord ratio
VLM	Vortex Lattice Method
W	Weight of the aircraft

1 Introduction

The aim of the Shockingly fast aircraft project was to develop a single seat, aerobatic electric powered aircraft to showcase both the benefits and the constraints provided by this type of propulsion. The design requirements were a maximum take off weight of $600kg$ and an endurance of 15 minutes [2]. The metrics used to judge the performance of the aircraft included the top speed and the time taken to perform a U-turn and a figure of 8. The design of an electric powered aircraft impose strict constraints on the aerodynamic design as the weight of the battery are highly dependent on the overall drag of the aircraft. Indeed the energy density of electric batteries is currently around 100 times lower than the one of gasoline [3].

This report will be mainly focused on the design of the wing. The design of the wing being closely linked to the overall design of the aircraft, it was important to develop methods so that new iterations could quickly be implemented and new parameters shared with the other teams. For this reason, the design was mainly realised using the vortex-lattice method (VLM) which was enhanced using a viscous model. The VLM is a powerful method to obtain estimates of the lift and drag of lifting surfaces at a low computational cost. In addition a simulation of the trajectory as a function of the aerodynamics characteristics of the aircraft was performed. These simulations showed the trade-offs required in the design of the aircraft to minimise the time taken to perform the figures. Finally near the end of the project winglets were designed to improve the overall performance of the aircraft.

2 Baseline configuration

In order to obtain a first sizing of the wing, a constraint diagram presented in figure 2 was done in collaboration with the propulsion and configuration team [4], [5] The following constraints were set on the design of the aircraft :

- A top speed of 130 ms^{-1}
- A sustained 8G turn, this constraint was later reduced to a 6 G turn due to an excessive thrust to weight requirement necessary to sustained an 8 G turn [4]
- A take off and landing distance of respectively 90 and 300 meters as set by the configuration team [6]

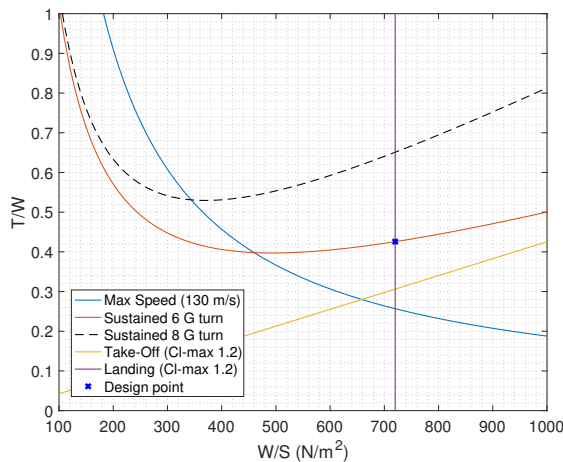


Figure 2: Constraint diagram

A wing loading (W/S) of 720 Nm^{-2} and a thrust to weight ratio (T/W) of 0.42 were chosen as the

design point.

For the first iteration, the weight of the aircraft was taken as the maximal takeoff allowed by the design briefing, that is $600kg$. Using this weight a reference area for the wing of 8.175 m was obtained.

At this stage of the design, the following variables and their implications were taken into account: the aspect ratio, the sweep and the taper ratio. The aspect ratio (AR) greatly influences the induced drag which can be written as:

$$C_{Di} = \frac{C_L^2}{e\pi AR} \quad (1)$$

where e is the Oswald efficiency being 1 for an elliptical loaded wing and less for a different lift distribution. As the aspect ratio is increased the tip vortices becomes weaker but the wing becomes more slender and the structural efficiency of the wing is reduced. In addition, the aspect ratio impacts the roll rate of the aircraft. As the AR is reduced, the center of gravity of the wing is brought closer to the fuselage reducing the overall second moment of inertia of the aircraft. Finally wing with higher AR are more prone to tip stall which is highly undesirable as this can cause an unexpected roll [7]. Following some literature research and looking at similar aerobatic aircrafts, it was decided to optimise the wing within an aspect ratio of 5 to 7 [8], [9].

The taper ratio (λ) can be used to reduce the amount of twist required to obtain an elliptical

lift distribution. A taper ratio close to 0.6 ensures the twist is minimised through the span. Adding some taper to a wing also increase the structural efficiency of the wing. Since the largest bending moment is experienced near the root, a wing with a larger taper ratio will have a larger chord at the root to withstand the bending moment. At the beginning of the project, it was chosen not to have any limitation on the taper ratio and see its impact on the maximal lift and drag. The sweep angle is mainly used to reduce the critical Mach number of transonic aircraft but as our maximum speed was set to 130ms^{-1} , which correspond to a Mach number of 0.38, the sweep angle was set to 0° .

Using the lifting line theory, it can be shown that an elliptical lift distribution over the span of the wing minimises the induced drag. For an electric aircraft a large fraction of the total weight is taken by the batteries and reducing the drag is therefore critical. Two design configurations resulting in an elliptical lift distribution were considered. The first design was an elliptical distribution of the chord. This solution offers more space near the root which would enable to store the batteries in the wing. Placing the batteries in the wing would relieve part of the load as the weight of the batteries would act in the opposite direction of the lift during high g-manoeuvre. However an elliptical chord distribution leads to an heavier structure is heavier and reduces the manoeuvrability of the aircraft [10], [11].

For the reasons mentioned above, it was chosen to obtain the elliptical lift distribution by adding some taper and twist to the wing. Using the lifting line theory, the angle required to obtain the elliptical lift distribution as a function of the chord is

$$\alpha(y) = \frac{C_L}{\pi AR} \left(\frac{8s\sqrt{1 - (y/s)^2}}{a_0 c(y)} \right) - \alpha_i \quad (2)$$

where y varies along the span of the wing from the center to the tip. The full derivation of equation (2) is presented in Appendix B. For a linear taper, the chord can be expressed as:

$$c(y) = c_{root} \left(1 - \frac{1 - \lambda}{s} y \right) \quad (3)$$

Combining equation (2) and 3, the twist distribution required to obtain an elliptical lift distribution as a function of the taper ratio can be calculated. The resulting twist distribution for a wing

with a reference area of 8.2m^2 and an AR of 6 is presented in figure 3. As the taper ratio is decreased below 0.6, the maximal twist angle moves towards the tip. This is highly undesirable as this means the tip of the wing will be more prone to stall.

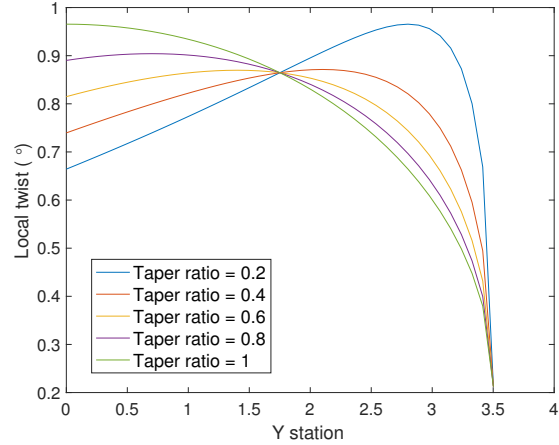


Figure 3: Local twist required for an elliptical lift distribution as a function of the taper ratio

3 Aerodynamic analysis method

3.1 XFOIL

XFOIL was used to test a range of aerofoils and obtain viscous data as well as their critical angle at subsonic speed. XFOIL is a widely used higher order panel coupled with viscous/inviscid interactions for the analysis of aerofoils [12]. The potential flow and boundary layers equations are solved simultaneously and therefore the separation point over the surface of the aerofoil can be predicted [13].

3.2 The Vortex-Lattice method

In the vortex lattice method (VLM), the 2D cross section are approximated by flat panels and the wing is divided into panels, each containing a horseshoe vortex. The strength of each vortex can be calculated by enforcing the impermeability condition at the collocation point of each panel. The influence of each panel on the other ones is then calculated resulting in both the span-wise and stream-wise variation of the lift. Once the wing loading is determined, the downwash velocity induced on each panel can be computed, and integrating the induced downwash over the wing, the overall induced angle of attack can be found. The VLM assumes that the flow is inviscid, irrotational, incompressible, that the lifting surfaces are thin and that the angle of attack is small [14].

Due to these assumptions only the induced drag and not the viscous one can be obtained using this method.

The Tornado implementation of the VLM in Matlab was used throughout this project [15]. This implementation was first compared with known results obtained from similar numerical methods [16] to validate the code. The Warren 12 planform, presented in Appendix C.1, was chosen for its sim-

ilarities to the wing designed in this project, that is a linear taper ratio, a low sweep angle and an intermediate AR. In addition, this validation procedure helped to determine which mesh distribution would yield the best results. The lift curve slopes, $a = \frac{\partial C_L}{\partial \alpha}$, obtained from Tornado using 6 panels chord-wise, 16 panels span-wise and the different mesh distributions were compared to the value of 2.743 unit/radian obtained in the literature [16]. The results are presented in table 1.

Table 1: Comparison of the results between Tornado and results from the literature

Mesh distribution	a (unit/radian)	Error (%)
Linear	2.7616	0.68
Spanwise half-cosine	2.7664	0.85
Spanwise half-cosine, chordwise cosine	2.7642	0.77
Spanwise cosine	2.7725	1.08

The number of panels used for the modelisation of the wing was found to have a very small impact on the convergence of the results to the value presented in the literature. For the rest of the project, the span half-cosine distribution was used as it provide the best estimation of the aerodynamic centre, which is a valuable information for the stability team [17].

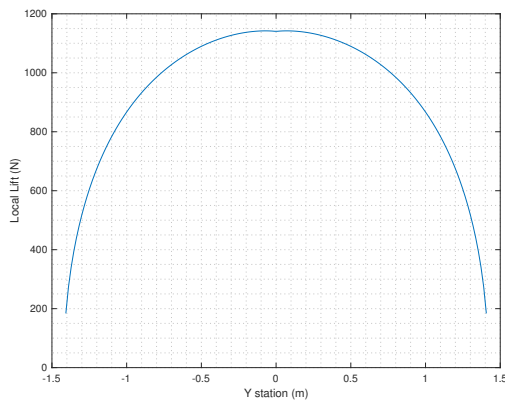
3.3 Viscous model

The vortex lattice method is an inviscid model and therefore only the induced drag is calculated. However as our aircraft flies for most of the trajectory at a low C_L , the induced drag is small in comparison with the viscous drag. For this reason, a viscous model was implemented in Matlab so that the drag and lift for a number of plan-

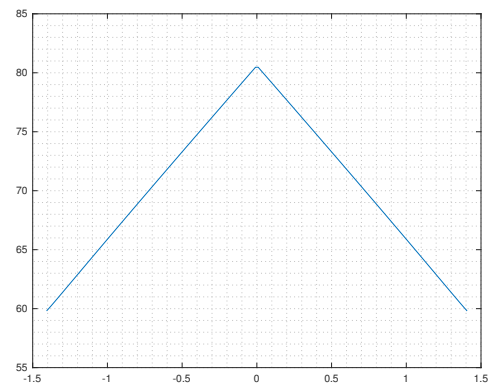
forms with different aerofoils could be obtained at a low computational cost. The lift distribution along the span was obtained using Tornado. Additionally, the 2D viscous characteristics of a given aerofoil were obtained using XFOIL at the Re based on the mean aerodynamic chord of the tapered wing. The drag from the 2D polar corresponding to the local lift was then interpolated to obtain a drag distribution along the span [18]. The total drag can then be obtained as follows:

$$C_D = \frac{1}{S_{ref}} \int_{-s}^s C_d(C_l(y)) c(y) + C_{D_i} \quad (4)$$

where C_d and C_l are respectively the 2D lift and drag coefficient. As the sweep angle of the wing is zero, the cross-flow is limited and therefore this method provides relatively accurate results.



(a) Local lift along the span of the wing



(b) Local drag along the span of the horizontal tail

Figure 4: local lift and drag along the span of the wing

This method was validated by comparing the results with data from an experiment performed in a wind-tunnel at a Mach number of 0.17 and a Re of 4.4×10^6 . The aerofoil used for this validation procedure is the NACA 64-210 and the wing geometry is presented in Appendix C.2. The lift and drag coefficient obtained from the viscous model and the VLM are compared to the results from the wind-tunnel in figure 5. An average error of 50.2% was observed between the drag obtained from Tornado and the experimental data. This error was reduced to 6.1% when the viscous effects were included in the calculations.

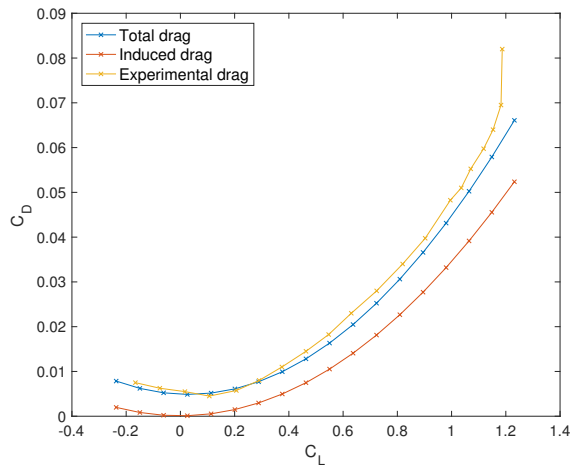


Figure 5: Comparison of the viscous model to the results obtained from the wind tunnel

The Matlab codes implementing the method de-

scribed in section 3.2 and 3.3 are presented in Appendix E. In addition, the distribution of the drag and lift along the span resulting from this method helped the structure team to optimise the thickness of the skin along the span of the wing [19], [20]. Both distributions are presented in figure 4.

3.4 Approximation of the critical angle and $C_{L_{max}}$

One of the main limitation of the vortex lattice method is that it does not take into account the boundary layer and as such no stall is predicted by this method. To obtain a first approximation of the $C_{L_{max}}$ of the wing, Tornado was run at increasing angle of attack with the designed wing until the maximal local lift coefficient, C_l , would be larger than the critical angle predicted by XFOIL for the given aerofoil. The code developed for this method is presented in E.4. It was later found using CFD simulations that the $C_{L_{max}}$ was overestimated using this method. The final results presenting the CFD simulations are detailed in section 5.3.

3.5 Drag buildup method

The drag build-up was used to obtain a first approximation of the parasite drag. This method is detailed in [11] and the parasite drag of the main components is presented in table 2. A zero-lift drag of 0.018 for the full aircraft was obtained and used for the rest of the design procedure.

Table 2: Parasite Drag Breakdown

	Wing	Fuselage	Horiz. Tailplane	Vert. Tailplane	Undercarriage	Total
C_{D_0}	0.006	0.007	0.002	0.001	0.002	0.018

4 Trajectory simulation

4.1 2D trajectory

In collaboration with the propulsion and configuration team, it was decided to simulate the trajectory using a Matlab script. The main metrics used to judge the performance of this project were a figure of 8 and a U-turn. The two figures were considered for the different constraints they impose on the aerodynamic characteristics of the aircrafts. For the U-turn, most of the trajectory is performed at the top speed and therefore the corresponding drag has to be reduced. For the figure of 8, the turning velocity is lower and it is therefore important that the $C_{L_{max}}$ of the wing is large

enough so that a 6 G turn can be performed at this speed.

At an early stage of the project, the trajectory was optimised in 2D, that is at a constant height. In this optimisation, it was assumed that the aircraft would decelerate before the turn and that the turn would be performed at a constant speed. Equations of motions and force equilibrium were used to model the acceleration and decelerations phase as presented in Appendix D.

The radius of the turn is constrained by the load factor and the turning velocity as follows.

$$R = \frac{V_2^2}{g\sqrt{n^2 - 1}} \quad (5)$$

Following the constraint diagram presented in section 2, a maximal load factor of 6 can be sustained through the turn.

4.2 3D trajectory

In a later stage of the project, a basic optimisation of the trajectory in 3D was made. It was expected that by transforming the kinetic energy to potential energy before turning, a reduction in the overall time taken to perform the two figures could be achieved. For this simulation, the accelerating and decelerating phase were performed while respectively losing and gaining altitude.

During the climb and descent phases, the climb rate ($\frac{dh}{dt}$) can be related to the accelerations as follows [21]:

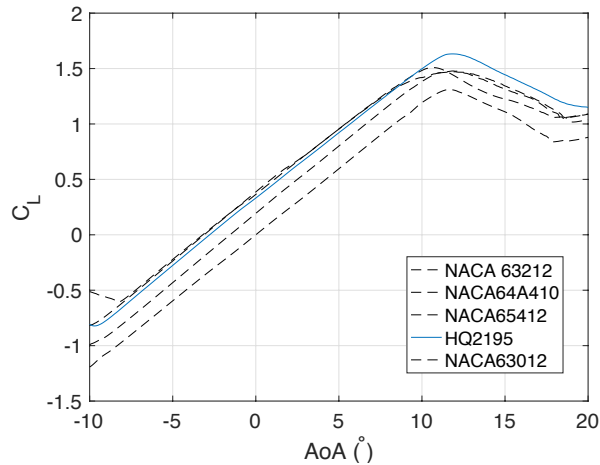
$$\frac{dh}{dt} = V \frac{T - D}{W} - \frac{V}{g} \frac{dV}{dt} \quad (6)$$

The specific excess power P_s was used to maximise the climb rate at each position during the climb or descent:

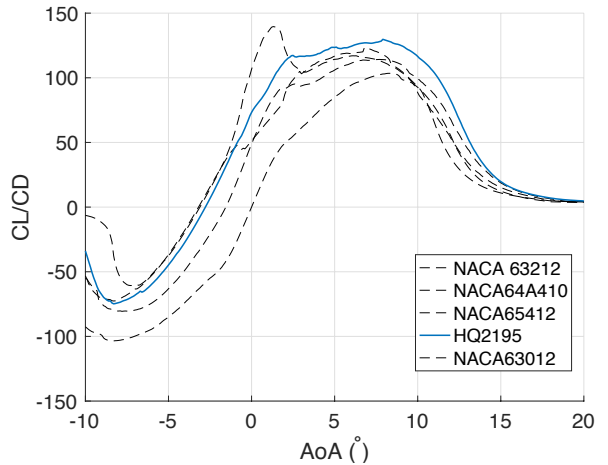
$$P_s = P_a - P_r = \frac{W}{g} V \frac{dV}{dt} + W \frac{dh}{dt} \quad (7)$$

where $P_a = T \times V$ is the available power and $P_r = D \times V$ is the required power. Figure 22 in Appendix D presents the contours of excess power as a function of the velocity and the altitude.

Once the optimal 3D trajectory was determined for both figures, the simulator was used to find the optimal aerofoil minimising the time taken to perform both figures as presented in section 5.3.



(a) Lift curve slope



(b) C_L/C_D vs α

Figure 7: Performance of the aerofoils selected for the first iterations

5 Wing planform and aerofoil selection

This section will present the different iteration of the wing which was refined throughout the project as the constraints imposed on the aerodynamics evolved.

5.1 First iteration

For the first iteration, it was important to quickly design a wing so that the other teams, mainly the control and propulsion team, would have necessary informations to start sizing their components. A first selection of aerofoil based on a high $C_{L_{max}}$, a high L/D at the C_L cruise of 0.21 and good stall properties was made. In order to evaluate the aerofoils, the Re was approximated as 7×10^6 based on an estimate of the average velocity over the U-turn of 118 m/s and a chord of 1.23m corresponding to a wing with an aspect ratio of 7 and no taper. The lift curve slope of the aerofoils selected as well as the lift to drag ratio are presented in figure 7.

As observed in figure 7, the HQ2195 aerofoil performed better than the other aerofoils for the criteria set above and was selected for the first iteration of the wing. The HQ2195 presented in figure 6 has a thickness to chord ratio (t/c) of 9.5% and a $C_{L_{max}}$ of 1.62

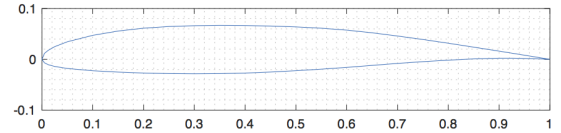


Figure 6: Profile of the HQ2195 aerofoil

Following the baseline configuration and as no constraint was yet imposed on the AR by the control team, an AR of 7 was selected in order to reduce the induced drag. The impact of the Aspect ratio on the induced drag as predicted by the VLM method is presented in figure 8.

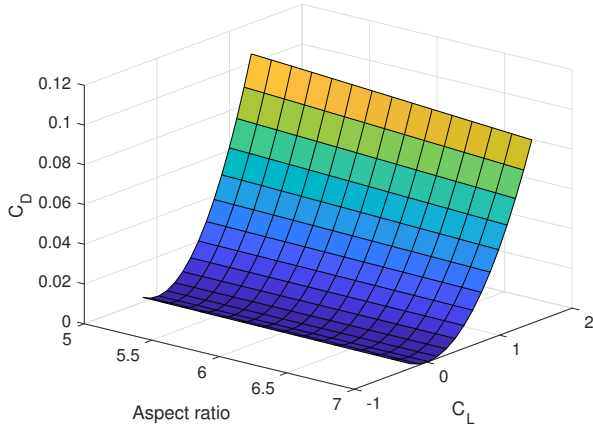


Figure 8: C_L and C_D vs. AR

As the elliptical lift distribution is enforced using twist, the induced drag predicted by the VLM method is independent of the taper ratio. Therefore for this iteration a taper ratio of 0.8 was chosen to improve the structural efficiency of the wing, while limiting the angle of twist at the tip required to enforce an elliptical lift distribution. The parameter of the wing designed during the first iteration are presented in table 3.

Table 3: Wing characteristics: First iteration

S_{ref} (m^2)	AR	Taper ratio	Span (m)	Root (m)	a (unit/radian)
8.2	7	0.8	7.56	1.18	4.73

In the first iteration, the $C_{L_{max}}$ was assumed to be 1.2, leading to a manoeuvring speed of 80 m s^{-1} for a 6 G turn. This speed was assumed as the turning velocity so that the turning radius would be minimised. The trajectory for the figure of 8 is presented in figure 21 in Appendix D.

5.2 Second iteration

For the second iteration, the maximal AR was reduced to 6 by the control team in order to achieve the desired roll rate [22]. The 3D simulation still being in development, the airfoil selected for the previous iteration was used and it was decided to maximise the $C_{L_{max}}$ of the wing in order to reduce the turning velocity in the 6G turns. This was done by varying the taper ratio and using the

method described in section 3.4. The results are presented in figure 9. A taper ratio of 0.44 was chosen as it provided a $C_{L_{max}}$ of 1.31. The main characteristics of the wing designed during the second iterations are presented in table 4.

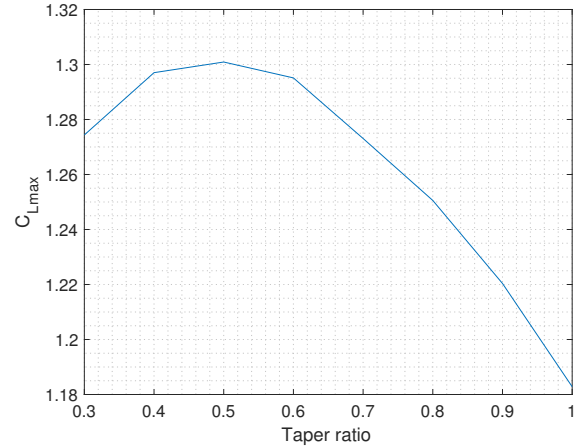


Figure 9: $C_{L_{max}}$ vs. Taper ratio

Table 4: Wing characteristics: Second iteration

S_{ref} (m^2)	AR	Taper ratio	Span (m)	Root (m)	a (unit/radian)
8.2	6	0.44	7	1.62	4.54

5.3 Third iteration

During the third iteration, all the tools described in section 3 were brought together in order to obtain an optimal design. As the viscous drag was added to the VLM model a new optimal taper ratio was obtained. As the elliptical lift distribution was enforced, the taper ratio had no impact on the induced drag. However the taper ratio influences the viscous drag as presented in figure 10. Figure 10 presents the sum of the induced and viscous drag as a function of the taper ratio. A taper ratio of 0.8 was selected as a compromise between the aerodynamic and structural efficiency of the wing [19].

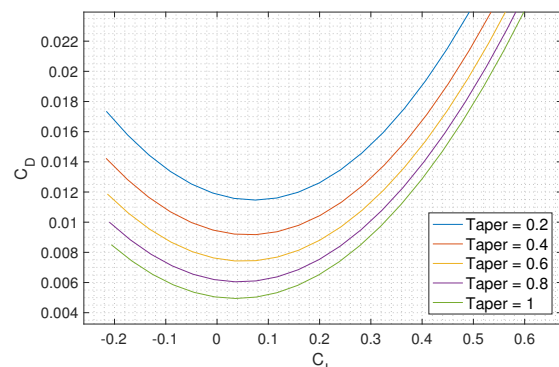


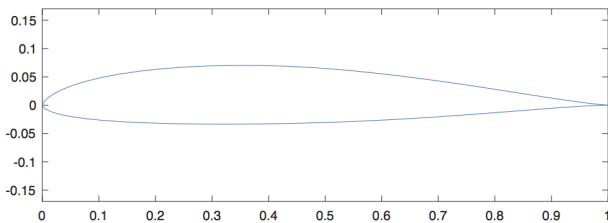
Figure 10: $C_{L_{max}}/C_D$ vs. Taper ratio

Table 5: Wing characteristics: Third iteration

S_{ref} (m^2)	AR	Taper ratio	Span (m)	Root (m)	a (unit/radian)
8.2	6	0.8	7	1.30	4.68

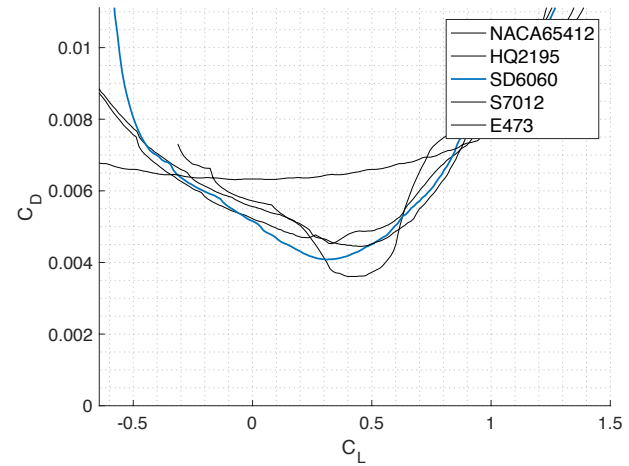
The final dimensions of the wing are presented in table 5. Once the development of the 3D trajectory code was finished, it was first used to determine the optimal turning velocity and the height which should be gained in order to minimise the time taken to slow the aircraft to the turning velocity. The 3D trajectory for the figure of 8 was found to be 7% faster than the 2D one. On the other hand, minimal improvement were observed for the U-turn as most of the trajectory is performed at the top speed. The optimal turning velocity for the U-turn and the figure of 8 were found to be respectively 111 ms^{-1} and 83.5 ms^{-1} . In order to perform a 6 G turn at 83.5 ms^{-1} , a C_L of 1.01 is required while the $C_{L_{max}}$ predicted for a taper ratio of 0.8 was of 1.27.

After the optimal trajectory was determined, the script simulating the trajectory was used with a new selection of aerofoils to find which aerofoil would minimise the time taken to execute the two figures (figure of 8 and U-turn). The aerofoils presented in figure 12 were selected for this iteration. For this selection, the $C_{L_{max}}$ was not a constraint anymore as the required C_L was found to be relatively low.

**Figure 11:** Profile of the SD6060 aerofoil

However as the speed varies over the whole trajectory, it was not possible to optimise the aerofoil

for a given C_L . A polar of C_D against C_L was obtained using the method described in section 3.3 for wings designed with each aerofoil selected for this iteration. The simulations of the two trajectories were then performed with each polar and the time taken to complete the two figures are presented in table 6. This process lead to the selection of the SD6060 presented in figure 11 as the main aerofoil for our wing. The breakdown of the time taken to perform the various section of the two figures are presented in section D.

**Figure 12:** C_D vs. C_L of the aerofoils selected for the third iteration

5.4 CFD results

CFD simulations of the wing, at a free stream velocity of 130 ms^{-1} , were performed near the end of the project. The objectives were to obtain more accurate data on the lift and drag as well as the critical angle and the $C_{L_{max}}$ of the wing. The mesh used for the CFD simulations is presented in Appendix section A.3 and more details on the simulations can be found in [11]. The results from the simulations along the ones obtained from Tornado are presented in figure 14. It can be observed that the lift curve slope obtained using the two methods are very similar at low angle of attack with an error of 8.67 % between 0° and 5° .

Table 6: Time taken to perform the figure of 8 and U-turn with the different aerofoil

aerofoil	U-turn	Δ	Figure of 8	Δ	$\sum \Delta$
SD 6060	314.99	0.00	45.09	0.36	0.36
NACA 65412	315.19	0.20	44.99	0.27	0.47
S7012	315.20	0.21	45.01	0.29	0.50
HQ2195	315.79	0.80	45.01	0.29	1.09
Eppler 473	316.59	1.60	44.72	0.00	1.60

This is consistent with the assumptions of the VLM which assumes small perturbations. CFD simulations showed a stall angle of 12° and a $C_{L_{max}}$ of 1.02, which is considerably lower than 1.27 as predicted using the method described in section 3.4. This $C_{L_{max}}$ is just above the C_L of 1.01 required to perform the 6G turn at 83.5 ms^{-1} . The lift to drag ratio provided by the viscous methods had a 31% error with respect to the results obtained from the CFD simulations. This error was not expected as the viscous method was

validated in section 3.3. However the difference could be due to the Mach number and compressibility effects. Indeed the VLM is an incompressible method but the compressibility effects were taken into account in the 2D drag polar extracted from XFOIL. On the other side, the CFD simulations were run with an incompressible model. Ultimately the solution to verify our results would be to test the wing in a wind tunnel so that both the CFD simulations and the viscous model could be validated at the given Mach numbers and Re.

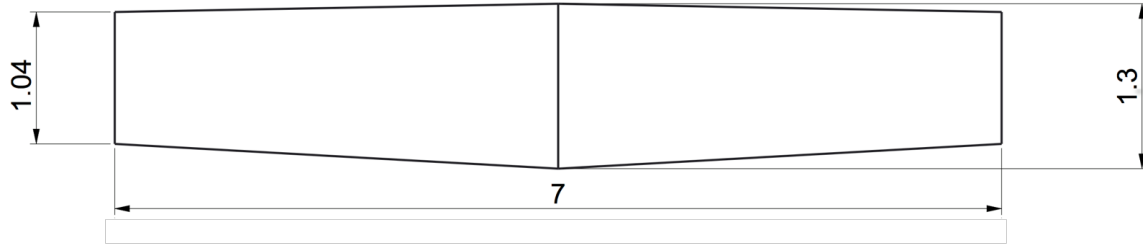
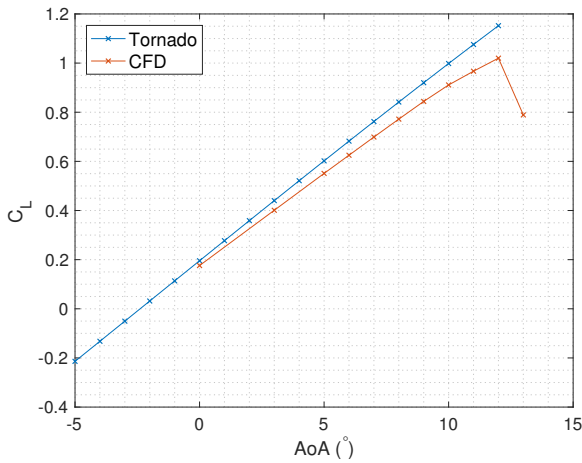
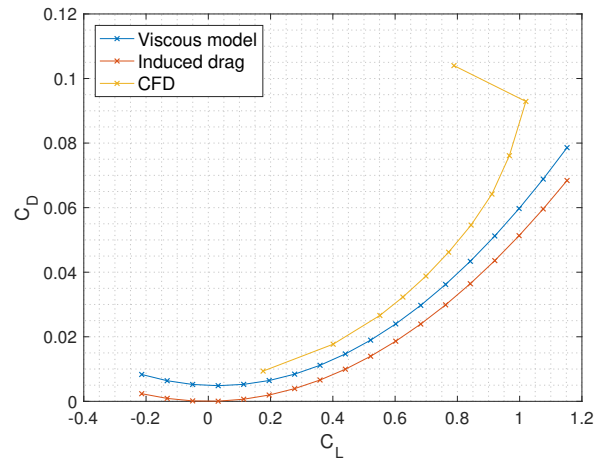


Figure 13: Third iteration: Wing dimensions



(a) Lift curve slope of the wing



(b) C_L/C_D vs α

Figure 14: Comparison of the results obtained from Tornado and CFD simulations

6 Aerodynamic analysis of the horizontal tail

The horizontal tailplane was designed by the control team. The lift and drag coefficient as well as their distribution along the span were then calculated following the procedure applied to the wing [23]. The results are presented in figure 15.

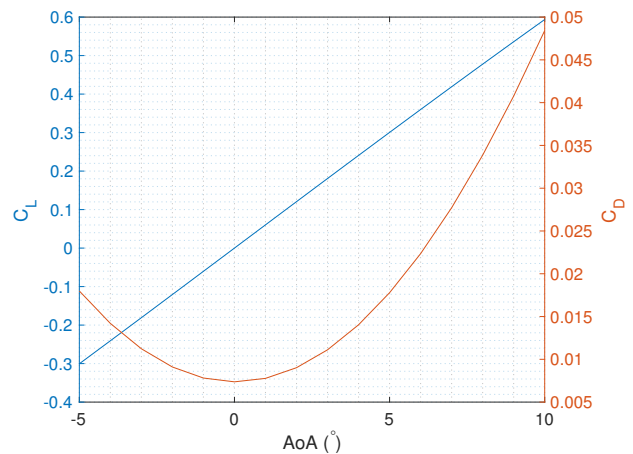


Figure 15: C_L and C_D of the horizontal tail vs AoA

6.1 Optimisation of the wing setting angle

Once the wing and the tail were designed, the wing and tail setting angle were left as free variables. These two angles were optimised so that the drag is minimised when the aircraft is trimmed at 130 ms^{-1} . The moment coefficient obtained from the VLM were used for this optimisation. To trim the aircraft the following constraints need to be satisfied [24]:

$$C_L = a_w(\alpha + i_w - \alpha_0) + \frac{S_h}{S_{ref}} a_h(\alpha + i_h)$$

$$= \frac{W}{0.5\rho S_{ref} V^2}$$

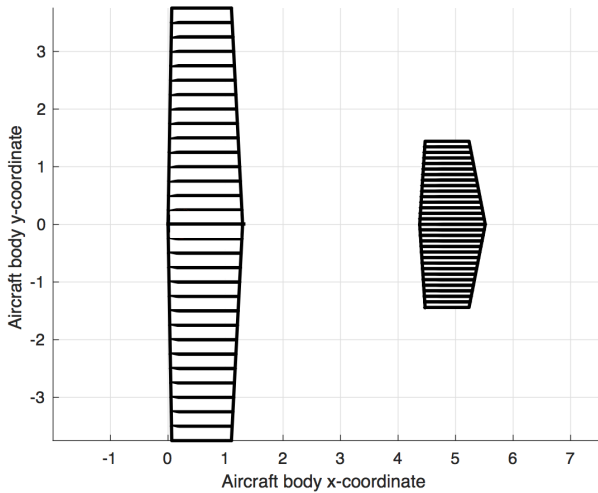
$$C_M = C_{M_{0w}} + \frac{x_{CG} - x_w}{\bar{c}} C_L - \frac{S_H l_h}{S_{ref} \bar{c}} C_{L_h} = 0$$

where i_w and i_h are respectively the setting angle of the wing and the horizontal tailplane. A combination of i_w and i_h satisfying these constraints

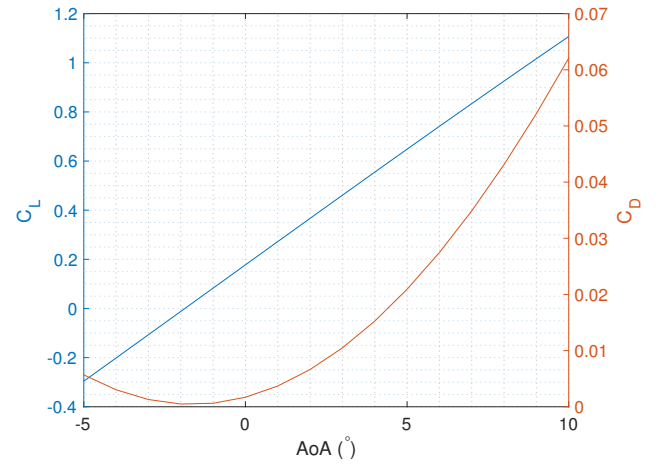
were tested and it was found that a wing and tail setting angle of respectively -2.17° and -0.55° would minimise the drag at 130 ms^{-1} .

7 Wing-Tail interactions

A simultaneous VLM analysis of the two surfaces was done in Tornado. The aim of this simulation was to estimate the influence of the downwash on the horizontal tailplane. As the horizontal tailplane is positioned on the same vertical plane as the wing, the downwash produced by the wing reduces the effective angle of the tailplane. When considering the two surfaces, a lift curve slope equals to 5.402 unit/radian was obtained. This is a reduction of 6% compared to the value obtained if the downwash of the wing on the tailplane is not taken into account. The geometry used in Tornado to represent the lifting surfaces as well as the C_L and C_D polar are presented in figure 16.



(a) Geometry of the lifting surfaces



(b) C_L and C_D vs α for the two lifting surfaces

Figure 16: Wing-tail interaction analysis

8 Wingtip devices

The main objective of fitting wingtip devices to a wing are to reduce the induced drag by increasing the circulation near the tip of the wing. Helmholtz theorem states that the strength of the vortex shed is equal to the change of circulation along the span [25]:

$$\frac{d\gamma_x}{dy} = -\frac{d\gamma_y}{dx} \quad (8)$$

If no wingtip devices are fitted to the wing, the pressure on the top and bottom surfaces have to be equal at the tip and therefore the circulation drops

to zero. By using wingtip devices, the circulation can be distributed in a more uniform manner along the span of the wing as the circulation does not drop to zero near the tip. The strength of the vortex shed and the drag are therefore reduced.

The addition of wingtip devices to our wing was motivated by two reasons. First by reducing the drag, the overall energy required to perform the figures would be decreased. However for this project the main motivation behind the winglets was to reduce the maximal power required from the battery in the 6G turn. Indeed the weight of the batteries (180 kg) is approximately propor-

tional to the maximal power which is

$$P = T \times V = D \times V$$

$$= (C_{D0} + C_{Di}) \frac{1}{2} \rho S_{ref} V^3$$

In general it is more efficient to increase the span of the wing and therefore the AR than including winglets in a new design [26]. However the constraints set by the control team would limit the AR of the wing to 6. Therefore a solution to

reduce the overall drag of the wing was to add wingtip devices.

At first both raked wingtip and winglet were considered. Raked wingtip are typically lighter than winglets. However they cause a larger bending moment and have a lower aerodynamic efficiency [1]. For these reasons it was chosen to prefer the winglets over the raked wingtip.

The design parameters presented in figure 17 need to be considered when designing winglets.

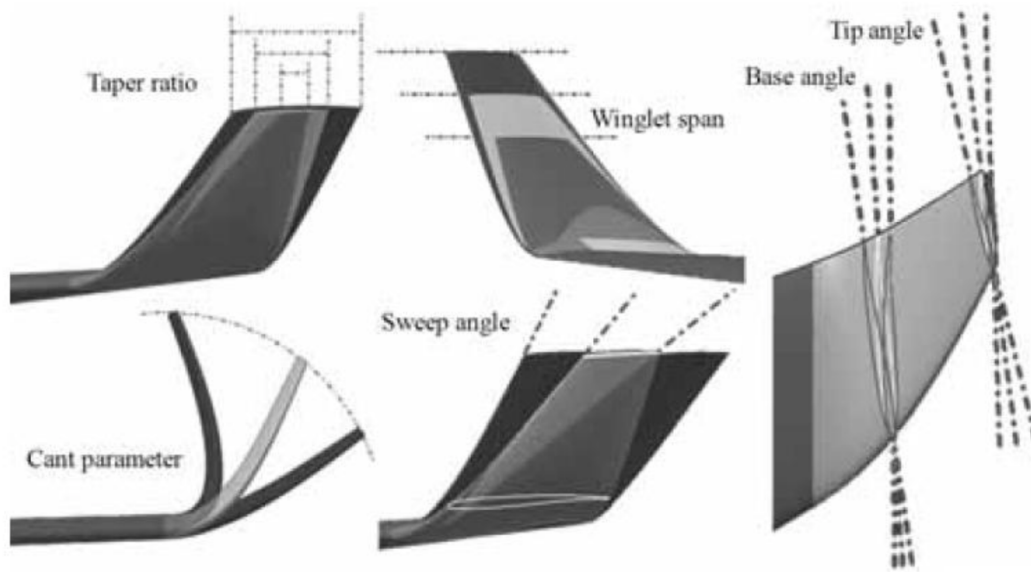
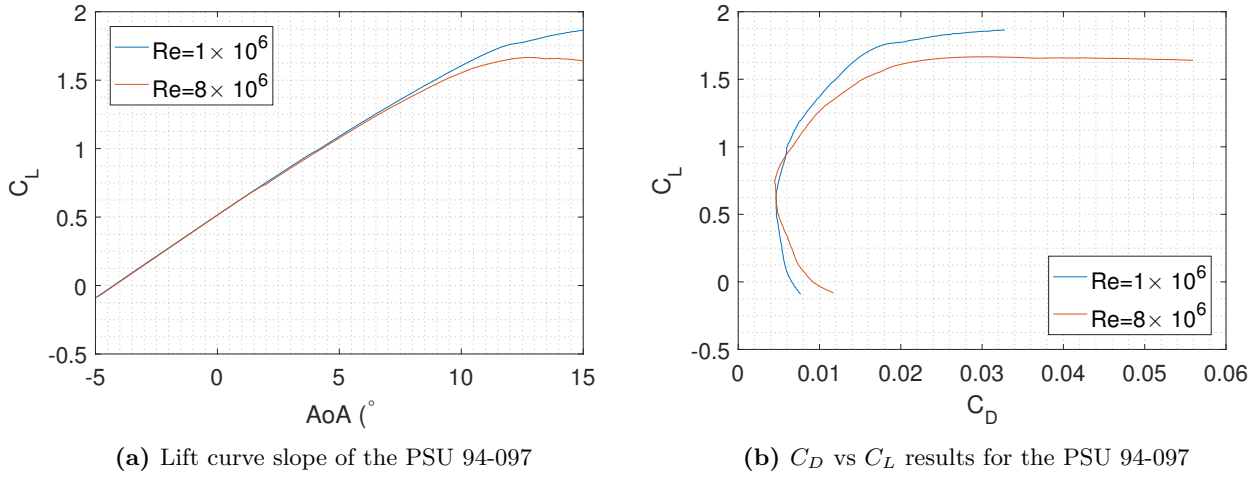
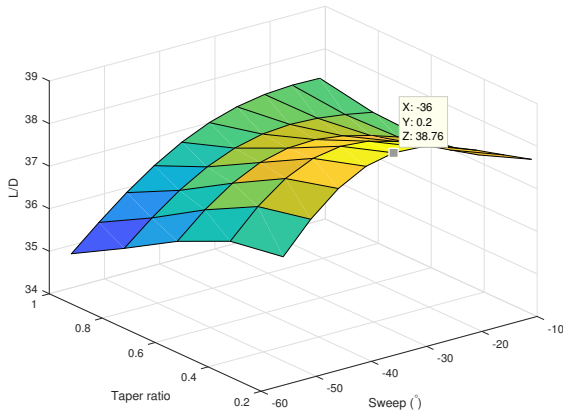


Figure 17: Design parameters of the winglet [1]

The aerofoil used for the wingtips has to provide a relatively large $C_{L_{max}}$ to avoid tip stall and to provide a high L/D over a large range of velocity. The PSU 94-097 aerofoil was chosen for the design of the winglets [27]. This aerofoil was designed by the University of Pennsylvania specifically for the design of winglets. It was initially designed to be used on winglets fitted to sailplanes. However the aerofoil was tested using XFOIL at a Re of 8×10^6 , corresponding to the mean chord of the winglet and was observed to retain the properties observed at lower Re as presented in figure 19/

From literature research, the height of the winglets was fixed to 10% of the semi-span, that is $0.35m$ [28]. The cant angle (angle between the vertical axis and the winglet) was set to 10° to make sure that as the wing experiences some deflections dur-

ing high G-manoeuve, the winglets do not produce downforce [27]. In addition, the wing span was reduced by 0.2 meter so that the reference area does not exceed the initial one by more than 4.5%. The taper ratio and the sweep were then optimised using the VLM, while the twist was varied linearly along the winglets to obtain good stall properties, that is a near constant repartition of the lift along the winglet. The VLM was observed to provide consistent results at low angle of attack, lower than 4° , and the winglets were therefore optimised at 3° . Following this optimisation presented in figure 19, a sweep angle of 38° and a taper ratio of 0.2 were chosen. The Matlab code used for this optimisation procedure is presented in Appendix E.6. For this design procedure the viscous drag was not included, hence the high values of the lift to drag ratio.

**Figure 18:** XFOIL results for the PSU 94-097**Figure 19:** L/D of the wing as a function of the sweep and taper ratio of the wingtip

In order to obtain a validation of the increased performances, CFD simulations were performed on the new wing with the added winglets. For these simulations, the mesh used on the simple wing was refined over the winglets. The mesh is presented in Appendix A.3 and more details on these simulations can be found [11]. Due to the computational cost of these simulations, only two of them were performed at an angle of attack of 0° and 8° . It was expected that at low angles of attack, the winglets would increase the overall drag due to the increase in reference area of the wing, but that they would reduce the induced drag at

higher angle of attack. As presented in table 7, at low angle of attack the increase in drag caused by the added area is compensated by the increase in lift due to the reduced downwash. At higher angle of attack, the wing showed both a decrease in the total drag as well as an increase in lift leading to an increase in L/D of 10% as presented in table 8. Looking at the differences in the streamline between the wings with and without winglets, it appears that the increase in L/D is mainly due to the winglets delaying the formation of the vortices at the wingtip. As observed in figure 20b, the tip vortices appear only aft of the wing and as such the flow over the wing is almost not affected by the tip vortices. This is different for the wing without winglets where the tip vortices move inwards and create a downwash over a relatively large portion of the wing as observed in figure 20a. In a further iteration of the winglets it would be interesting to extend the winglets below the wing, as it can be noted that the vortices are created by the flow coming from the lower surface of the wing. Also CFD simulations including the horizontal tailplane could help to calculate the effects of the wingtips on the downwash experienced by the horizontal tailplane. Indeed the tip vortices on the wing with the winglets seems to be concentrated near the tip of the wing and the downwash on the tail should therefore be reduced.

Table 7: Performance of the winglets at 0°

	Without winglets	With winglets
C_L	0.176	0.191
C_D	9.37×10^{-3}	1.01×10^{-2}
L/D	18.8	18.8

Table 8: Performance of the winglet at 8°

	Without winglets	With winglets
C_L	0.772	0.823
C_D	4.62×10^{-2}	4.47×10^{-2}
L/D	16.7	18.4

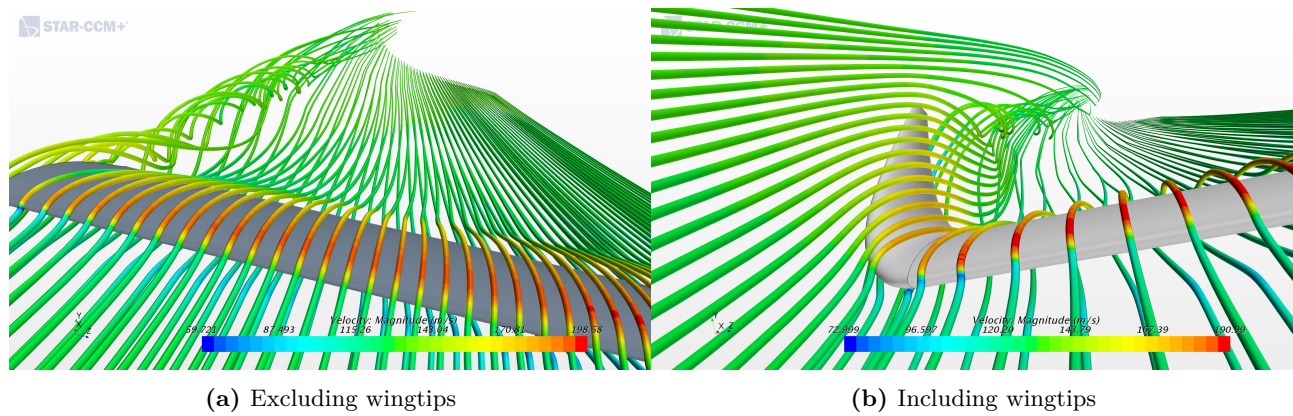


Figure 20: Comparison of the flow over the wing with and without winglets at 8°

9 Conclusion

The objectives of this report were to present the design process of the wings and the implications of the electric propulsion on the aerodynamic team. Throughout the project, models of low computational power were developed so that new designs could be iterated quickly. The viscous model was validated with wind tunnel data within a small error, but was later found to have an average of 31% with the CFD simulations. The error in the CFD simulations could be due to the mesh type and physics model used for the simulations but wind tunnel testing would be required to confirm this hypothesis. Wind tunnel testing would also bring relevant information on the overall parasitic drag. Concerning the simulations of the trajectory, improvement could be obtained by reducing the velocity while turning. Encouraging results have been observed in 2D and the same procedure in 3D could lead to a significant reduction of the overall time taken to perform the two figures.

The winglets showed an encouraging improvement of 10% in the L/D at an angle of attack of 8° , while no reduction was present in the L/D at lower angles of attack. The impact on the structure due to the addition of wingtips devices were quickly considered when choosing winglets over raked wing tips. However in a further iteration these considerations would have to be refined taking into account the aerodynamic efficiency as well as the structural efficiency.

Overall the design of the wing was successful as the objectives of the aircraft, that is a top speed of 130 ms^{-1} and a sustained 6 G turn were achieved. This project showed the potential of using electrical propulsion over conventional propulsion and applying the recommendations above, this type of propulsion definitely needs to be considered in the design of the future generation of aerobatic aircraft.

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Appendix A Additional figures

A.1 2D trajectory for the figure of 8

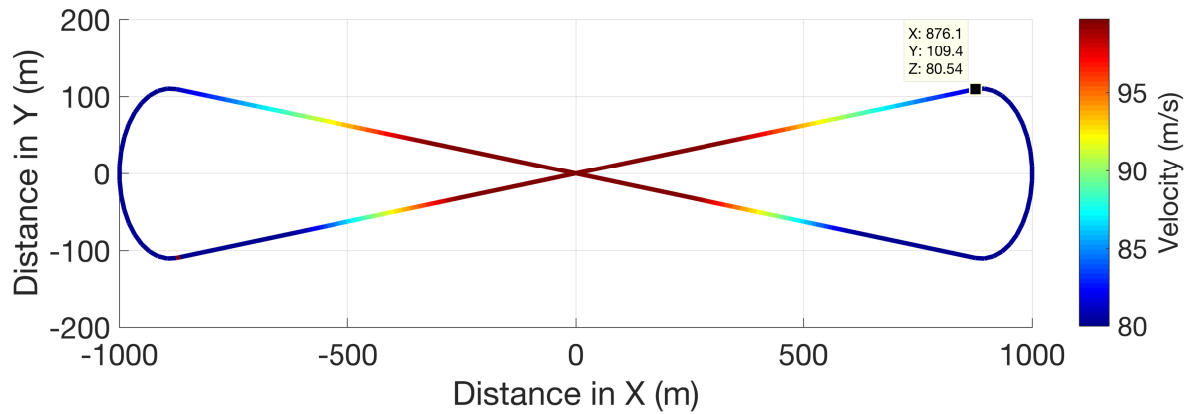


Figure 21: Plot of the 2D trajectory

A.2 Excess power

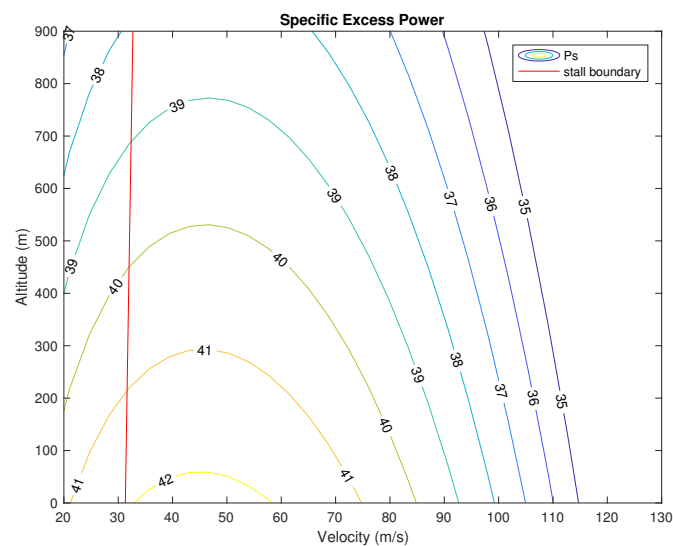


Figure 22: Contour of excess power as a function of the altitude and the flight speed

A.3 Mesh used for the CFD simulations

Figure 23 and 24 shows the mesh used for the simulation of the wing without winglets. Figure 25 shows the refinement of the mesh used for the wing with the winglets.

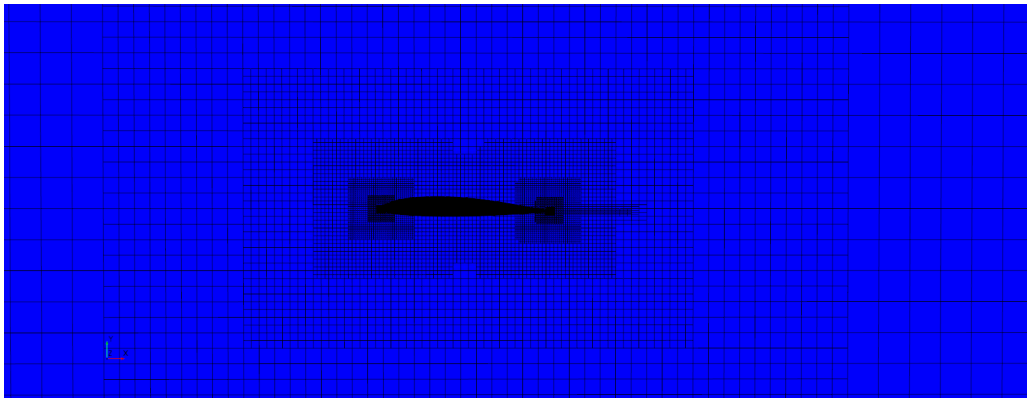


Figure 23: Wake refinement

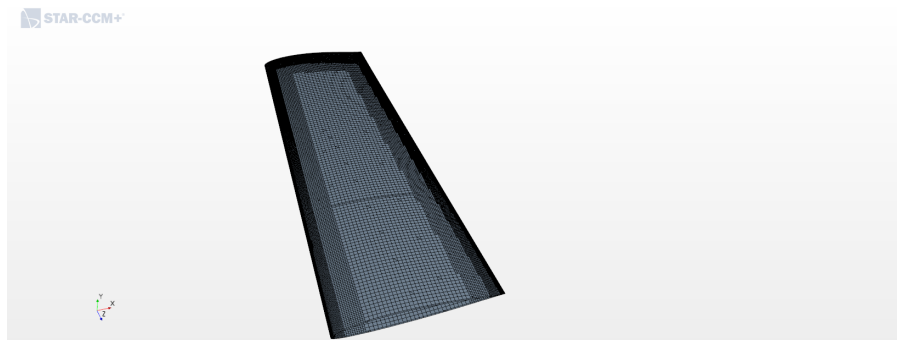


Figure 24: Trailing and leading edge refinement

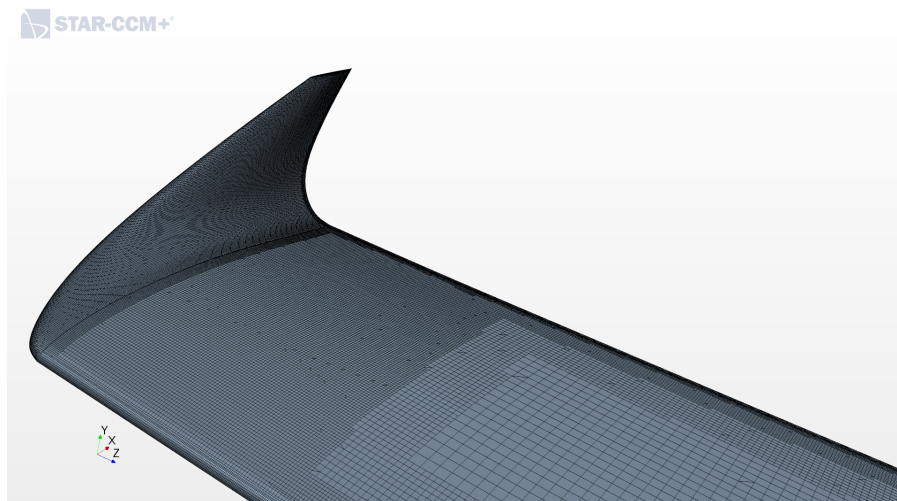


Figure 25: Mesh refinement at the wingtip for the wing including the winglets

A.4 Local lift and drag

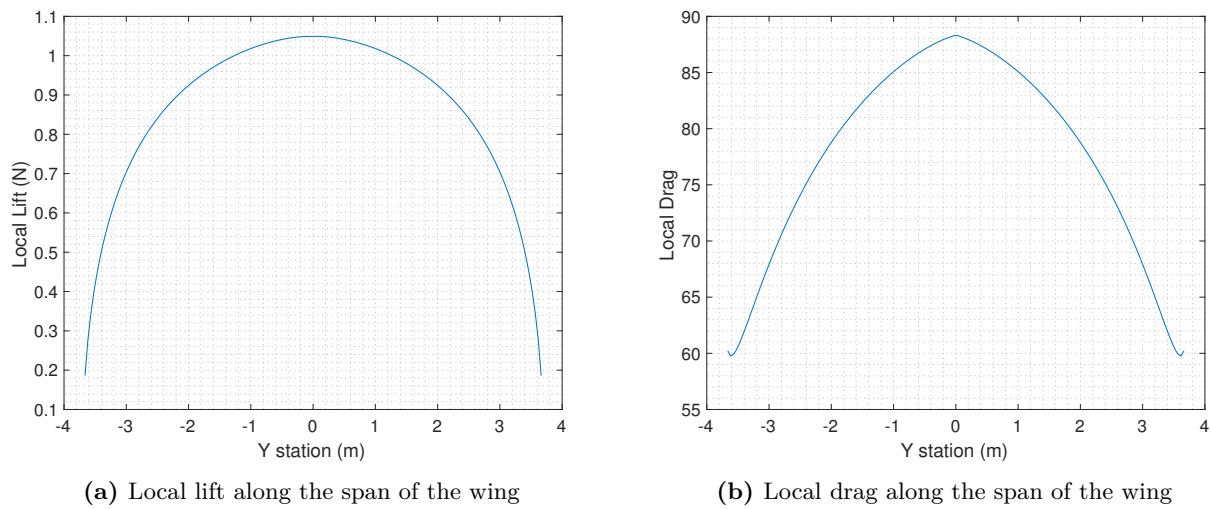


Figure 26: local lift and drag along the span of the horizontal tail

Appendix B Elliptical lift distribution

In the collocation solution method, the loading is expressed as a Fourier series along the span of the wing:

$$\Gamma(\theta) = 4sU_\infty \sum_{n=1}^{\infty} A_n \sin(n\theta) \quad (9)$$

For an elliptic loading, only the first coefficient is non-zero

$$L = \int_{-s}^s \rho U \Gamma dy = \frac{\pi}{2} \rho U \Gamma_0 s \quad (10)$$

where Γ_0 is the maximal circulation. The lift coefficient can be rewritten as:

$$C_L = \frac{\pi/2 \rho U \Gamma_0 s}{1/2 \rho U^2 S_{ref}} \quad (11)$$

In addition the circulation can be written as

$$\Gamma = \frac{1}{2} U c(y) a_0 [\alpha(y) + \alpha_i(y)] \quad (12)$$

or as only A_1 is non-zero it is equal to:

$$\Gamma = \Gamma_0 \left(1 - \left| \frac{y^2}{s^2} \right| \right)^{\frac{1}{2}} \quad (13)$$

where α_i is the induced angle of attack which is constant along the span for an elliptical loading and is equal to:

$$\alpha_i = -\frac{C_L}{\pi AR} \quad (14)$$

Equating equation (12) and equation (13) we obtain:

$$\alpha(y) = \frac{C_L}{\pi AR} \left(\frac{8s \sqrt{1 - (y/s)^2}}{a_0 c(y)} \right) - \alpha_i \quad (15)$$

Appendix C Model validation: Wing geometry

C.1 Warren planform

The warren planform as defined in [16] is presented in figure 27 and the main wing characteristics are presented in table 9.

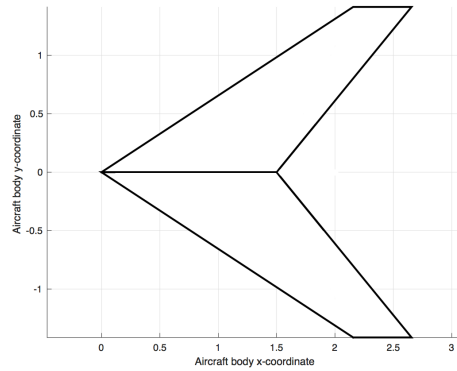


Figure 27: Warren planform

Table 9: Warren 12: planform geometry

S_{ref} (m^2)	AR	Λ_{LE} ($^\circ$)	λ	c_{root} (m)
2.83	2.83	53.54	$1/3$	1.5

C.2 Viscous model validation

The planform used to validate the viscous model with the wind tunnel data.

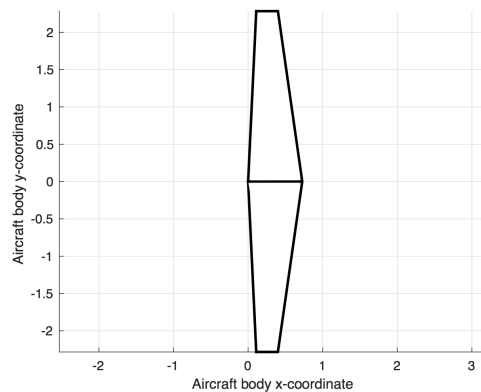


Figure 28: Planform used for validation of the viscous model

Table 10: Planform used for the validation of the viscous model

S_{ref} (m^2)	AR	Λ_{LE} ($^\circ$)	λ	Washout ($^\circ$)	Dihedral ($^\circ$)	c_{root} (cm)
2.32	9	0	0.4	2	3	72.6

Appendix D Trajectory: Additional informations

The 3D trajectory for the figure as well as the the gain in altitude is presented in figure 30 The acceleration and deceleration were modeled using the equation of motions and force equilibrium as follows:

$$\begin{bmatrix} x \\ \dot{x} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad (16)$$

$$\begin{bmatrix} \dot{x} \\ \ddot{x} \end{bmatrix} = \begin{bmatrix} x_2 \\ \frac{T-D}{m} \end{bmatrix} \quad (17)$$

where x_1 is the displacement, x_2 the velocity and T is the thrust (T) obtained from the thrust to weight ratio ($T/W=0.42$) giving a maximal thrust of $2472N$. The drag is composed of the parasite drag and the induced drag so that $C_D = C_{D_0} + C_{D_i}$.

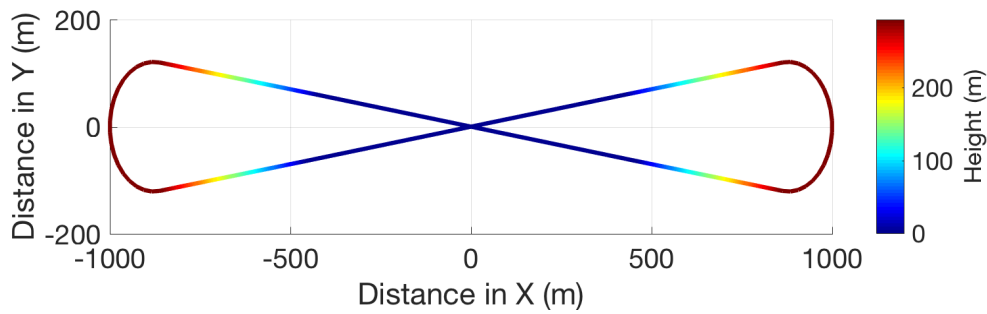


Figure 29: 3D trajectory for the figure of 8

The time taken to perform the U-turn as a function of the turning velocity is presented in

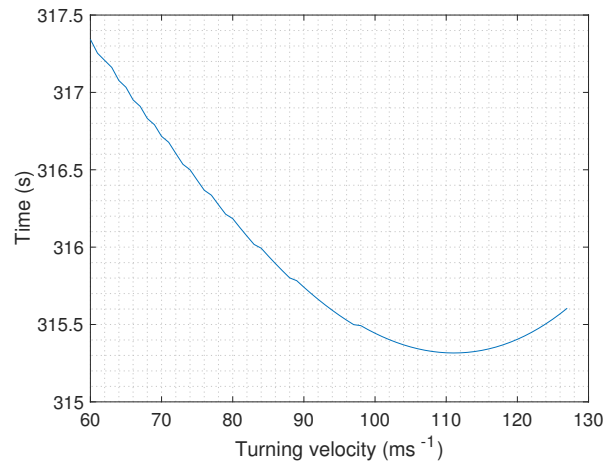


Figure 30: Time taken to perform the U-turn as a function of the turning speed

The breakdown of the time taken to perform the various section of the figure of 8 and the U-turn are presented in table 11 and 12.

Table 11: Breakdown of the U-turn

Section	Velocity (ms^{-1})	Time (s)
Straight 1	130	151.1
Climb	130 to 111	3.1
Turn	111	6
Descent	111 to 129	3
Acceleration	129 to 130	0.7
Straight2	130	151.0
Total	129	315.0

Table 12: Breakdown of the figure of 8

Section	Velocity (ms^{-1})	Time (s)
Straight	130	2.1
Climb	130 to 83.5	7.7
Turn	83.5	5
Descent	83.5 to 139.7	7.7
Total ($\times 2$)	103	45.1

Appendix E Matlab scripts

E.1 Definition of the twist

This script is used in Appendix E.3

```
function [y,c,alphat,V,s,Clc]=funtwist(Ar,gamma)
%Script written by H.Turbe, 17.05.2017
%function to compute the required twist for elliptical distribution
%input Ar, gamma, S, Clc,
%output, y,chord (array),alpha(array),V,s,Clc
    global geo
    global Aircraft
    global state

%Vmd=(K./Aircraft.Cd0).^0.25*sqrt(2*Aircraft.WS); %Vmin drag
V=130;

Clc=Aircraft.WS./(0.5*state.rho.*V.^2); %lift coefficient at given speed

s=sqrt(Aircraft.S*Ar/4); %semi-span
y=[0:s/(geo.nelem):s];
gamma0=Clc*V*Aircraft.S/(pi*s); %circulation mid-wing
alphai=-Clc/(pi*Ar); %induced angle of attack
geo.c=4*s/(Ar*(1+gamma)); %root chord
c=geo.c*(1-(1-gamma)/s*y); %chord over the span
alphat=Clc/(pi*Ar)*(8*s/Aircraft.a0*sqrt(1-(y./s).^2)./c)-alphai; %required twisting angle
end
```

E.2 Geometry definition

This script is used in Appendix E.3

```
function [latticetype] = definition-setup(Aerofoil,c,alpha,s)
%Script written by H.Turbe, 17.05.2017
global Aircraft
global state
global geo
latticetype = 1; %Standard VLM lattice
%% Geometry definition %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
geo.nwing= 1; %Number of wings in the design,
% determines the number of ROWS in the
% geometry variables

%geo.nelem= 10; %Number of partitions in the design,
% Determines the number of COLUMNS in
% the geometry variables

geo.ref_point= [0.25,0,0]; %Position of the reference point in
% the aircraft coordinate system
% Moments are taken around this point

geo.CG= [0.25,0,0]; %Position of the center of gravity in
% the aircraft coordinate system.
% Rotations are made about this point

%geo.c= 1; %Root chord of each wing

for i=1:geo.nelem(1)
geo.b(1,i)= s/(geo.nelem(1)-1) ; %Semispan of each partition of each wing
geo.dihed(1,i)=0; %Dihedral of each partition of each wing
%[rad]
```

```

geo.T(1,i)=      c(i+1)/c(i);  %Taper ratio of each partition.

geo.SW(1,i)=      0*pi/180; %c/4 sweep of each partition. [rad]

geo.TW(1,i,1)=    alpha(i);  %Twist of partition inboard section
geo.TW(1,i,2)=    alpha(i+1); %Twist of partition outboard section

geo.nx(1,i)=      4; %Number of panels in the X direction of each section

geo.ny(1,i)=      4; %Number of panels in the Y direction of eachs section

geo.flapped(1,i)=  0;          % 0 if no flap

geo.fnx(1,i)=      0;  %Number of panels in the X direction of  %the flap of each partiton

geo.fsym(1,i)=      0; %Control surface deflect symmetrically bit

geo.fc(1,i)=        0; %Part of chord that is flapped of  %each partition

geo.flap_vector(1,i)= 0; %Flap deflection, [rad], right hand positive outboard

geo.foil(1,i,1)= Aerofoil;
geo.foil(1,i,2)= Aerofoil;

geo.meshtype(1,i)=  1;
end                %Type of mesh to be used

geo.symmetric=      1;          %Symmetry bit for each wing
geo.startx=          0;          %X coordinate of each wing apex
geo.starty=          0;          %Y coordinate of each wing apex
geo.startz=          0;          %Z coordinate of each wing apex

%% End Aero structure definition %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% Define the state %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% state.AS=          118;          %Airspeed m/s
state.alpha=         15*pi/180;    %Angle of attack, radians
state.betha=          0;          %Angle of sideslip, radians
state.P=              0;          %Rollrate, rad/s
state.Q=              0;          %pitchrate, rad/s
state.R=              0;          %yawrate, rad/s
state.adot=           0;          %Alpha time derivative rad/s
state.bdot=           0;          %Betha time derivative rad/s
state.ALT=            50;          %Altitude, m

[T, ah, Ph, rho] = atmosisa(state.ALT);
state.rho=          rho;          %Desity, kg/m^3
state.pgcorr=       0;          %To Apply prandtl glauert compressibility
                           %correction

```

E.3 VLM and viscous model

```

%Script written by H.Turbe, 1.06.2017
%script to create polar using Tornado and viscous model
clc
clear
close all

global path
path={'/Users/hturbe/OneDrive - Imperial College London/ecole/Imperial/3rd year/GDP/tornado2'};
% add the path for the tornado code - you will have to modify this line

```

```

addpath(genpath(path{1,1}));
global Aircraft
global state
global geo
geo = struct([]);
geo(1).useless=1;

Aircraft.W=600*9.81; %weight aircraft
Aircraft.WS=720; %wing loading
Aircraft.S=Aircraft.W/Aircraft.WS;
alpha= [-5:1:12]*pi/180; %range angle of attack in radians for polar

state.ALT= 50; %altitude
[T, ah, Ph, rho] = atmosisa(state.ALT); %used for to determine Mach number and density
state.rho=rho;

geo.nelem= 30; %Number of partitions in the design,
taper=0.8; %taper ratio
Ar=6;

Aerofoil = {'sd6060.DAT'}; %airfoil coordinates
airfoil=load('sd6060.2.mat'); %2D lift and drag
airfoil=airfoil.sd6060;

Aircraft.a0=6.63; %2D lift curve slope required for elliptical lift distribution
[y,c,alphan,Vmd,s,Clc]=funtwist(Ar,taper); %twist distribution
state.AS= 130; %Airspeed m/s
pos1=find(airfoil.Alpha==-4); %position in fit to avoid stall
pos2=find(airfoil.Alpha== 7.9);
cdfit=polyfit(airfoil.CL(pos1:pos2),airfoil.CD(pos1:pos2),2); %fit CD vs. CL

[latticetype] = definition_setup(Aerofoil,c,alphan,s); %create wing geometry

state.Mach=state.AS/ah; %Computing mach number
for i=1:length(alpha)
    state.alpha=alpha(i);
%% Generate the Lattice
[lattice,ref]=fLattice_setup2(geo,state,latticetype); %generate lattice

%% Compute coefficients of the wing
[results.CD0 results.Re results.Swet results.Vol]=zeroliftdragpred(state.Mach,state.ALT,geo,ref);

[results]=solver9(results,state,geo,lattice,ref);
[results]=coeff_create3(results,lattice,state,ref,geo);
results.CL_CD = results.CL/results.CD;
CL(i)=results.CL %store CL for each AoA
CDtor(i)=results.CD %store induced drag for each AoA
c2=geo.c*(1-(1-taper)/s.*results.ystation(length(results.ystation)/2+1:end));
c3=[fliplr(c2'),c2']; %chord variation for local lift
for n=1:length(results.ystation)
    cd_loc(n)=polyval(cdfit,full(results.CL_local(n)))*c3(n); %evaluate 2d drag for given local CL
end
CD(i)=1/Aircraft.S*trapz(results.ystation,cd_loc) %integrate drag distribution for total viscous drag
CDtot(i)=CD(i)+results.CD %total drag
CM(i)=results.Cm; %store Cm for each AoA
end
CL=[alpha.*180/pi;CL]
CDtot=[alpha.*180/pi;CDtot]
CM=[alpha.*180/pi;CM]
cd(['/Users/hturbe/OneDrive - Imperial College London/ecole/Imperial/3rd year/GDP');
save('polar_val_sd6060.mat','CL','CDtot','CM')

```

E.4 Estimation of the $C_{L_{max}}$

```
%script to determine max CL of the wing using Tornado
```



```

%Script written by H.Turbe, 25.05.2017
clc
clear
close all
global path
path={'/Users/hturbe/OneDrive - Imperial College London/ecole/Imperial/3rd year/GDP/tornado2'};
addpath(genpath(path{1,1}));
global Aircraft
global state
global geo
geo = struct([]);
geo(1).useless=1;

Aircraft.W=600*9.81; %Weight of the aircraft
Aircraft.WS=720; %wing loading first iteration
Aircraft.S=Aircraft.W/Aircraft.WS;
Aircraft.a0=6.6436; %2D lift curve slope required for elliptical lift distribution
cl_max=1.402; %max sectional lift coefficient for sd6060

state.ALT= 50; %altitude (m)
[T, ah, Ph, rho] = atmosisa(state.ALT);
state.rho=rho;

geo.nelem= 30; %Number of partitions in the design,
Aerofoil = {'sd6060.DAT'};
taper=[0.2:0.2:1];
Ar=6;
for i=1:length(taper)
    state.alpha= 10*pi/180;
    [y,c,alphat,V,s,Clc]=funtwist(Ar,taper(i));
    results.CL_local=0;
    [latticetype] = definition_setup(Aerofoil,c,alphat,s); %define wing geometry
    state.AS= V; %Airspeed m/s
    while max(results.CL_local)<=cl_max %check if max local Cl is greater than stall angle airfoil
        state.alpha=state.alpha+0.1*pi/180 %increment alpha

        %% Auxillary computations
        state.Mach=state.AS/ah; %Computing mach number

        %% Generate the Lattice
        [lattice,ref]=fLattice.setup2(geo,state,latticetype);

        %% Compute coefficients of the wing
        [results.CD0 results.Re results.Swet results.Vol]=zeroliftdragpred(state.Mach,state.ALT,geo,ref);

        [results]=solver9(results,state,geo,lattice,ref);
        [results]=coeff_create3(results,lattice,state,ref,geo);
    cd('/Users/hturbe/OneDrive - Imperial College London/ecole/Imperial/3rd year/GDP/tornado')
    end
    alpha(i)=state.alpha*180/pi;
    CL_max(i)=results.CL; %max 3D CL
end
disp(state.alpha(end)*180/pi)
disp(results.CL(end))

```

E.5 Geometry of the wing and the winglets

```

%script written by H.Turbe to design the winglets 12.06.2017
function [latticetype] = definition_setup_wingtip(Aerofoil,c,alpha,s,elem.wing,elem.wingtip,sweep.tip)

global Aircraft
global state
global geo
latticetype = 1; %Standard VLM lattice
%% Geometry definition %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```

```

geo.nwing=      1;           %Number of wings in the design,
                           %determines the number of ROWS in the
                           %geometry variables

%geo.nelem=      10;         %Number of partitions in the design,
                           %Determines the number of COLUMNS in
                           %the geometry variables

geo.ref_point=   [0.25,0,0]; %Position of the reference point in
                           %the aircraft coordinate system
                           %Moments are taken around this point

geo.CG=          [0.25,0,0]; %Position of the center of gravity in
                           %the aircraft coordinate system.
                           %Rotations are made about this point

%geo.c=          1;         %Root chord of each wing

for i=1:elem_wing
geo.b(1,i)= s/(elem_wing-1) ; %Semispan of each partition of each wing
geo.dihed(1,i)=0;           %Dihedral of each partition of each wing
                           %[rad]

geo.T(1,i)=      c(i+1)/c(i); %Taper ratio of each partition.

geo.SW(1,i)=      4.23*pi/180; %c/4 sweep of each partition. [rad]

geo.TW(1,i,1)=   alpha(i);  %Twist of partition inboard section
geo.TW(1,i,2)=   alpha(i+1); %Twist of partition outboard section

geo.nx(1,i)=      4; %Number of panels in the X direction of each section
geo.ny(1,i)=      4; %Number of panels in the Y direction of eachs section

geo.flapped(1,i)= 0;        % 0 if no flap

geo.fnx(1,i)=      0; %Number of panels in the X direction of %the flap of each partiton

geo.fsym(1,i)=     0; %Control surface deflect symmetrically bit

geo.fc(1,i)=       0; %Part of chord that is flapped of %each partition

geo.flap_vector(1,i)= 0; %Flap deflection, [rad], right hand positive outboard

geo.foil(1,i,1)= Aerofoil;
geo.foil(1,i,2)= Aerofoil;

geo.meshtype(1,i)= 2; %Type of mesh to be used
end
%% definition geometry wingtip curved part

R=0.2; %radius blend wing and winglet
h=0.35; %height winglet
iter=floor(elem_wingtip*R/h);
incl=10; % cant angle
x=linspace(0,R,iter);
y=-sqrt(R^2-x.^2)+R;
chord=linspace(c(end),taper_tip*c(end),elem_wingtip); %start taper winglet

for n=1:length(chord)-1
    taper_li(n)=chord(n+1)/chord(n);
end

geo.T(1,elem_wing+1 :elem_wing+ elem_wingtip-1)=taper_li; %define taper

```

```

twist=linspace(alpha(elem_wing),alpha(elem_wing)+twist_tip,elem_wingtip);
geo.TW(1,elem_wing+1 :elem_wing+ elem_wingtip-1,1)=twist(1:end-1); %define twist
geo.TW(1,elem_wing+1 :elem_wing+ elem_wingtip-1,2)=twist(2:end);
for i=elem_wing+1 : elem_wing+ iter-1

geo.b(1,i)= x(i-elem_wing+1)-x(i-elem_wing) ; %Semispan of each partition of each wing
geo.dihed(1,i)=atan((y(i-elem_wing+1)-y(i-elem_wing))/(x(i-elem_wing+1)-x(i-elem_wing))); %Dihedral of each partition
%geo.dihed(1,i)=30*pi/180;
                                % [rad]

%geo.T(1,i)= taper_li(i); %Taper ratio of each partition.

geo.SW(1,i)= sweep_tip; %c/4 sweep of each partition. [rad]

% geo.TW(1,i,1)= alpha(elem_wing); %Twist of partition inboard section
% geo.TW(1,i,2)= alpha(elem_wing); %Twist of partition outboard section

geo.nx(1,i)= 4; %Number of panels in the X direction of each section
geo.ny(1,i)= 4; %Number of panels in the Y direction of each section

geo.flapped(1,i)= 0; % 0 if no flap

geo.fnx(1,i)= 0; %Number of panels in the X direction of the flap of each partition
geo.fsym(1,i)= 0; %Control surface deflect symmetrically bit
geo.fc(1,i)= 0; %Part of chord that is flapped of each partition

geo.flap_vector(1,i)= 0; %Flap deflection, [rad], right hand positive outboard

geo.foil(1,i,1)= {'psu94.DAT'}; % Airfoil used for the winglet
geo.foil(1,i,2)= {'psu94.DAT'};

geo.meshtype(1,i)= 2;

geo.symmetric= 1; %Symmetry bit for each wing
geo.startx= 0; %X coordinate of each wing apex
geo.starty= 0; %Y coordinate of each wing apex
geo.startz= 0; %Z coordinate of each wing apex
end

%% definition geometry wingtip straight part
chord=linspace(c(end),taper_tip*c(end),elem_wingtip-(iter-1));

%geo.TW(1,elem_wing+iter :elem_wing+ elem_wingtip-1,1)=twist(1:end-1);
%geo.TW(1,elem_wing+iter :elem_wing+ elem_wingtip-1,2)=twist(2:end);
height=(h-R)*2/(1+taper_tip); %height section used to calculate dihedral
for i=elem_wing+iter :elem_wing+ elem_wingtip-1

geo.b(1,i)= height/(elem_wingtip-1-iter) ; %Semispan of each partition of each wing
geo.dihed(1,i)=(90-incli)*pi/180; %Dihedral of each partition of each wing
%geo.dihed(1,i)=5*pi/180;
                                % [rad]

%geo.T(1,i)= chord(i-(elem_wing+iter)+2)/chord(i-(elem_wing+iter)+1); %Taper ratio of each partition
%geo.T(1,i)= taper_li(i);

geo.SW(1,i)= sweep_tip; %c/4 sweep of each partition. [rad]

% geo.TW(1,i,1)= twist(i-(elem_wing+iter)+1)*pi/180; %Twist of partition inboard section
% geo.TW(1,i,2)= twist(i-(elem_wing+iter))*pi/180; %Twist of partition outboard section

```

```

geo.nx(1,i)=      4; %Number of panels in the X direction of each section

geo.ny(1,i)=      4; %Number of panels in the Y direction of eachs section

geo.flapped(1,i)=  0;          % 0 if no flap

geo.fnx(1,i)=      0;  %Number of panels in the X direction of  %the flap of each partiton

geo.fsym(1,i)=      0; %Control surface deflect symmetrically bit

geo.fc(1,i)=        0; %Part of chord that is flapped of  %each partition

geo.flap_vector(1,i)= 0; %Flap deflection, [rad], right hand positive outboard
%Aerofoil_tip = {'sd6060.DAT'};
geo.foil(1,i,1)= {'psu94.DAT'};
geo.foil(1,i,2)= {'psu94.DAT'};

geo.meshtype(1,i)=  2;

geo.symmetric=     1;          %Symmetry bit for each wing
geo.startx=        0;          %X coordinate of each wing apex
geo.starty=        0;          %Y coordinate of each wing apex
geo.startz=        0;          %Z coordinate of each wing apex
end
%% End Aero structure definition %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%% Define the state %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% state.AS=        118;          %Airspeed m/s
%state.alpha=      15*pi/180;    %Angle of attack, radians
state.betha=       0;          %Angle of sideslip, radians
state.P=          0;          %Rollrate, rad/s
state.Q=          0;          %pitchrate, rad/s
state.R=          0;          %yawrate, rad/s
state.adot=       0;          %Alpha time derivative rad/s
state.bdot=       0;          %Betha time derivative rad/s
state.ALT=        50;          %Altitude, m

[T, ah, Ph, rho] = atmosisa(state.ALT);
state.rho=        rho;          %Desity, kg/m^3
state.pgcorr=     1;          %Apply prandtl glauert compressibility
                           %correction

```

E.6 Design of the winglet

```

%script written by H.Turbe to design the winglets 12.06.2017
clc
clear
close all
global path
path={'/Users/hturbe/OneDrive - Imperial College London/ecole/Imperial/3rd year/GDP/tornado2'};

addpath(genpath(path{1,1}));
global Aircraft
global state
global geo
geo = struct([]);
geo(1).useless=1;

Aircraft.W=600*9.81;
Aircraft.WS=720; %wing loading first iteration
Aircraft.S=Aircraft.W/Aircraft.WS;
Aircraft.Cd0=0.0179; %Cd0 to be changed only first iteration
Aircraft.a0=6.63; %from XFRLR5 required for elliptical lift distribution

```

```

%Aircraft.a0=8.0993; %for e473
%Aircraft.a0=8.0993;

state.alpha=      3*pi/180; %AoA in radian
state.ALT=      50;
[T, ah, Ph, rho] = atmosisa(state.ALT);
state.rho=rho;

%geo.c=      1;      %Root chord of each wing
elem_wing=20; %elements used for the wing
elem_wingtip=20; %elements used for the wingtip
geo.nelem= elem_wing+ elem_wingtip-1 ; %Number of partitions in the design,
taper=0.8; %taper of the wing
Ar=6; %AR of the wing
[y,c,alphat,Vmd,s,Clc]=funtwist_wingtip(Ar,taper,elem_wing); %define twist wing
state.AS=      130;      %Airspeed m/s
Aerofoil = {'sd6060.DAT'}; %load airfoil wing, airfoil tip set definition.setup_wingtip
sweep_tip=[-45:5:0]*pi/180; %sweep winglet
taper_tip=[0.2:0.2:1]; %taper ratio winglet
twist_tip=-2*pi/180; %linear twist variation winglet
for i=1:length(sweep_tip)
    for j=1:length(taper_tip)
        [latticetype] = definition_setup_wingtip(Aerofoil,c,alphat,s-0.3,elem_wing,elem_wingtip,sweep_tip,taper_tip,twist_tip);

        %% Auxillary computations
        [T, ah, Ph, rho] = atmosisa(state.ALT); %Calling International Standard atmosphere.

        state.Mach=state.AS/ah; %Computing mach number

        %% Generate the Lattice
        [lattice,ref]=fLattice.setup2(geo,state,latticetype);

        %% Compute coefficients of the wing
        [results.CD0 results.Re results.Swet results.Vol]=zeroliftdragpred(state.Mach,state.ALT,geo,

        [results]=solver9(results,state,geo,lattice,ref);
        [results]=coeff_create3(results,lattice,state,ref,geo);
        Sref(i,j)=ref.Sref %reference area wing
        CL(i,j)= results.CL; %lift coefficient
        CD(i,j)= results.CD; %drag coefficient
        CL_CD(i,j) = results.CL/results.CD
        cd(path{1,1})
        cd('/Users/hturbe/OneDrive - Imperial College London/ecole/Imperial/3rd year/GDP/tornado');
    end
end
[X,Y] = meshgrid(-45:5:0,0.2:0.2:1);
surf(X,Y,CL_CD) %plot results
set(gca,'FontSize',15)
xlabel('Sweep (^circ)')
ylabel('Taper ratio')
zlabel('L/D')

```

E.7 Trajectory U-turn

```

%script written by H.Turbe to determine the trajectory of the u-turn
%15.0.6.2015
clc
clear
close all
load('polar_val_sd6060.mat') %load polar airfoil use for the drag of the wing
alpha= [-5:1:10];
global Cl_fit
Cl_fit=polyfit(CL(1,:),CL(2,:),1); %fit for CL vs Alpha
global Cd_fit
Cd_fit=polyfit(CDtot(1,:),CDtot(2,:),2); %fit for CD vs Alpha

```

```

global Aircraft
global env
%% define aircraft variables
Aircraft.Cd0=0.019; %parasite drag aircraft
Aircraft.m=600;
Aircraft.W=Aircraft.m*9.81;

env.rho=1.225;
env.g=9.81;
env.n=6; %max load factor
Aircraft.S=Aircraft.W/Aircraft.WS; %area wing
Aircraft.TW=0.47;
Aircraft.Tmax=Aircraft.TW*Aircraft.W;
Aircraft.Vmax=130;
%Vturn=sqrt(2*env.n*Aircraft.WS/(env.rho*Aircraft.Clmax));
Vturn=111; %turning velocity obtained from optimisation
Rturn=Vturn^2/(env.g*sqrt(env.n^2-1)); %turn radius

%% climb to lose speed
Aircraft.alt=50;
xclim=ones(2,2)*10^5; %create empty array for ode45
t2=2; %start iteration climb
Aircraft.power =0;
while xclim(end,2)>Vturn %climb until turning speed is reached
[tclimb,xclim] = ode45(@climbfun,[0 t2],[0,Aircraft.Vmax]); %solve equation of motions
t2=t2+0.1; %increment time climb
end
gamma=asind(Aircraft.dhdt./xclim(:,2)); %angle of climb
dclimb=0; %horizontal distance during climb
for i=2:length(tclimb)
    deltax=xclim(i,1)-xclim(i-1,1);
    deltad=deltax*cosd(gamma(i-1));
    dclimb=dclimb+deltad;
    dist(i)=dclimb; %horizontal distance travelled during climb
end

%% descent after turn
hgain=Aircraft.dhdt*tclimb(end); %total gain altitude
Aircraft.alt=-1; %set descent in func ode45
t2=0.5;
Aircraft.power =0; %Without thrust
while Aircraft.dhdt*t2<hgain
[t descent,x descent] = ode45(@climbfun,[0 t2],[0,Vturn]);
t2=t2+0.1;
end

gammad=asind(Aircraft.dhdt./xdescent(:,2));
d descent=0; %horizontal distance during climb
for i=2:length(tclimb)
    deltax=xclim(i,1)-xclim(i-1,1);
    deltad=deltax*cosd(gamma(i-1));
    d descent=d descent+deltad;
end

%% accleration after turn and descent
xacc=zeros(2,2);
x0 = [0 xdescent(end,2)];
Aircraft.power=1; %With thrust
t2=0.5;
while xacc(end,2)<Aircraft.Vmax
tarray = [0 t2];
[tacc,xacc] = ode45(@accelerate,tarray,x0);
t2=t2+0.2;
end

```

```

t_turn=pi*Rturn/Vturn; %time to perform the turn
t_straight1=(20*10^3-dclimb)/Aircraft.Vmax; %time to perform the first straight
t_straight2=(20*10^3-xdescent(end,1))/Aircraft.Vmax; %time to perform the second straight

t_tot=t_straight1+tclimb(end)+t_turn+tdescent(end)+tacc(end)+t_straight2; %total time
figure(2)
y=tclimb*Aircraft.dhdt;
plot(dist,y)

%% function for the ode45
function dvdt=climbfun(t,x)
global Aircraft
global env
global Cl_fit
global Cd_fit
if Aircraft.power ==1 %set power on or off depending climb or descent
    T=Aircraft.Tmax;
else if Aircraft.power==0;
    T=0;
    end
end

if Aircraft.alt== -1 %climb or descent
    dhdt=-1.5*Aircraft.dhdt;
else
    dhdt=Aircraft.dhdt;
end
dvdt = zeros(2,1);
n=1; %load factor in climb
CL = 2*n*Aircraft.m*env.g./(env.rho*x(2)^2*Aircraft.S); %compute required CL
alpha=(CL-Cl_fit(2))/Cl_fit(1); %corresponding alpha to CL
CDi = polyval(Cd_fit,alpha); %extract CD wing from VLM+viscous method
D = 0.5*env.rho*x(2)^2*Aircraft.S*(Aircraft.Cd0+CDi); %Generate drag force
dvdt(1)=x(2);
dvdt(2)=( (T-D)*x(2)/Aircraft.W-dhdt)*env.g/x(2);
end

function dxdt = accelerate(t,x)
global env
global Aircraft
global Cl_fit
global Cd_fit
if Aircraft.power ==1
    T=Aircraft.Tmax;
else if Aircraft.power==0;
    T=0;
    end
end
dxdt = zeros(2,1);
CL = 2*Aircraft.m*env.g./(env.rho*x(2)^2*Aircraft.S);
alpha=(CL-Cl_fit(2))/Cl_fit(1);
CDi = polyval(Cd_fit,alpha);
%CDi=0;
D = 0.5*env.rho*x(2)^2*Aircraft.S*(Aircraft.Cd0+CDi); %Generate drag force

dxdt(1) = x(2);
dxdt(2) = (T-D)/Aircraft.m;
end

```